Why complex numbers arise

Real rotation matrix

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

has no nonzero real eigenvector when $\theta \not\equiv 0 \pmod{\pi}$. Its eigenvalues are

$$\lambda = e^{\pm i\theta}.$$

Complex eigenvectors describe oscillations, roots of unity, and periodic signals. A real matrix has conjugate-paired complex eigenvalues, and real solutions combine the corresponding modes with conjugate coefficients.

We use $i^2 = -1$. Electrical engineering often writes j for the same imaginary unit.

Arithmetic in the complex field

Write $z = a + bi$ with real a, b. Add and subtract components.

$$(a + bi) + (c + di) = (a + c) + (b + d)i.$$

Multiplication gives

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

The complex numbers form a field. For nonzero $a + bi$,

$$\frac{1}{a + bi} = \frac{a - bi}{a^2 + b^2}.$$

Multiplication commutes, and the positive squared modulus of a nonzero number supplies the reciprocal denominator. Standard basis vectors also work in $\mathbb{C}^n$, with complex coefficients.

Conjugation, modulus, and dimension

Conjugate:

$$\overline{a+bi} = a - bi.$$

Modulus:

$$|z| = \sqrt{z\overline{z}} = \sqrt{a^2+b^2}.$$

Conjugation preserves sums and products.

$$\overline{z+w} = \overline{z} + \overline{w}, \qquad \overline{zw} = \overline{z}\,\overline{w}.$$

Also $|zw| = |z||w|$, and the triangle inequality holds.

The complex dimension of $\mathbb{C}^n$ is n. Its real dimension is $2n$, because every component has a real and imaginary part. Specify the scalar field when counting dimensions or ranks.

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Polar form and Euler's formula

A nonzero complex number

$$z = a + bi$$

has

$$r = |z|, \qquad \theta = \operatorname{atan2}(b, a),$$
$$z = re^{i\theta} = r(\cos\theta + i\sin\theta).$$

Euler formula:

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

Multiplication multiplies moduli and adds arguments.

$$(re^{i\theta})(se^{i\phi}) = rse^{i(\theta+\phi)}.$$

Powers:

$$(re^{i\theta})^k = r^k e^{ik\theta}.$$

For example,

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Polar form and Euler's formula (continued)

$$1 + i = \sqrt{2} e^{i\pi/4},$$

$$(1 + i)^8 = 16 e^{2\pi i} = 16.$$

Arguments differing by 2π represent the same number. A branch choice matters for multivalued operations such as the complex logarithm.

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Roots of unity

Solve

$$z^n = 1.$$

Set

$$\omega = e^{2\pi i/n}.$$

The roots are

$$1, \omega, \omega^2, \dots, \omega^{n-1}.$$

They are equally spaced on the unit circle, forming a regular polygon for $n \geq 3$.

If $z = re^{i\theta}$ and $z^n = 1$, then

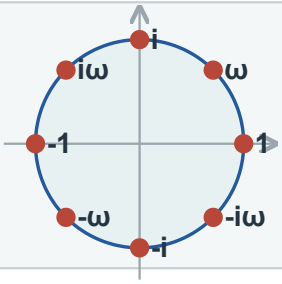
$$r = 1, \qquad n\theta = 2\pi k.$$

For $n = 4$, the roots are $1, i, -1, -i$. They also generate Fourier vectors and eigenvectors of cyclic shifts.

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Roots of unity (continued)

Eighth roots of unity: equally spaced on the unit circle



ω=e^(2πi/8)

roots：1,ω,…,ω⁷

$F_8F_8^H=8I$

Orthogonal columns; Fourier basis

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The complex inner product

Our convention is

$$\langle u, v\rangle = u^H v = \sum_j \overline{u_j} v_j,$$

where

$$u^H = \overline{u^T}.$$

For $u = (1, i)^T$,

$$\|u\|^2 = u^H u = |1|^2 + |i|^2 = 2,$$

rather than the bilinear expression $1^2 + i^2 = 0$.

Inner product properties:

$$\langle u, v\rangle = \overline{\langle v, u\rangle},$$
$$\langle u, v + w\rangle = \langle u, v\rangle + \langle u, w\rangle,$$
$$\langle u, \alpha v\rangle = \alpha\langle u, v\rangle,$$

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The complex inner product (continued)

$$\langle \alpha u, v \rangle = \overline{\alpha} \langle u, v \rangle.$$

Conjugation belongs to the first argument throughout. A different convention requires consistent changes in every formula.

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Orthogonality and the four subspaces

Orthogonality means

$$u^H v = 0.$$

For example,

$$u = (1, i)^T, \qquad v = (1, -i)^T,$$
$$u^H v = (1, -i)(1, -i)^T = 1 + (-i)(-i) = 1 - 1 = 0.$$

The complex four-subspace relations use the conjugate transpose.

$$C(A) \perp N(A^H), \qquad C(A^H) \perp N(A).$$

Ordinary A^T does not generally give the Hermitian orthogonal complements. For instance, $(A^H y)^H x = y^H A x$, which vanishes when $Ax = 0$.

Dimensions:

$$\dim C(A) = \dim C(A^H) = r,$$
$$\dim N(A) = n - r, \qquad \dim N(A^H) = m - r.$$

Hermitian matrices

A square matrix is Hermitian if

$$S^H = S.$$

Entrywise:

$$s_{ji} = \overline{s_{ij}}.$$

Its diagonal entries are real. For every z,

$$(z^H S z)^H = z^H S^H z = z^H S z.$$

The scalar equals its own conjugate, so

$$z^H S z \in \mathbb{R}.$$

For example,

$$S = \begin{bmatrix} 2 & 1+i \\ 1-i & 3 \end{bmatrix}$$

is Hermitian.

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Hermitian eigenvalues and eigenvectors

If nonzero z satisfies

$$Sz = \lambda z,$$

multiply on the left by z^H.

$$z^H S z = \lambda z^H z.$$

The left side is real and $z^H z > 0$, proving λ real.

If

$$Su = \lambda u, \qquad Sv = \mu v,$$

then, using reality of λ,

$$\lambda u^H v = (Su)^H v = u^H S v = \mu u^H v.$$

For distinct eigenvalues,

$$(\lambda - \mu) u^H v = 0,$$

so the eigenvectors are orthogonal.

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Hermitian eigenvalues and eigenvectors (continued)

For completeness of the eigenbasis, maximize the real-valued quadratic form $z^H S z$ on the complex unit sphere. Stationarity in real and imaginary directions gives $Sq = \lambda q$. The complement $q^\perp$ is invariant because $q^H S v = (Sq)^H v = 0$. Restriction remains Hermitian, so induction yields a complete orthonormal eigenbasis. Gram-Schmidt can be applied within repeated eigenspaces. Thus

$$S = Q \Lambda Q^H,$$

where $Q^H Q = I$ and Λ is real diagonal.

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Unitary matrices

A square matrix $Q \in \mathbb{C}^{n \times n}$ is unitary if

$$Q^H Q = I.$$

It also satisfies $QQ^H = I$, so

$$Q^{-1} = Q^H.$$

It preserves inner products.

$$\langle Qu, Qv \rangle = (Qu)^H (Qv) = u^H v.$$

Thus $\|Qu\| = \|u\|$.

If $Qu = \lambda u$,

$$\|u\| = \|Qu\| = |\lambda| \|u\|,$$

so

$$|\lambda| = 1.$$

The determinant also records length preservation.

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Unitary matrices (continued)

$$1 = \det(Q^H Q) = (\det Q)^H \det Q = |\det Q|^2,$$

Hence $|\det Q| = 1$. Real orthogonal matrices are the real special case of unitary matrices.

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Normal matrices

A square matrix is normal if

$$AA^H = A^H A.$$

A complex square matrix is unitarily diagonalizable exactly when it is normal. For sufficiency, take a unit eigenvector and extend it to a unitary basis, producing $\begin{bmatrix} \lambda & b^H \\ 0 & C \end{bmatrix}$. Equality of the $(1,1)$ entries of AA^H and $A^H A$ gives $|\lambda|^2 + \|b\|^2 = |\lambda|^2$, so $b = 0$. The remaining block is normal, and induction proves the result. Necessity follows by multiplying $A = Q\Lambda Q^H$ and its adjoint. Important classes include

- Hermitian:eigenvalues real;
- skew-Hermitian $K^H = -K$:eigenvalues pure imaginary;
- unitary:eigenvalues modulus 1.

Ordinary diagonalizability requires an eigenbasis. Unitary diagonalizability requires an orthonormal eigenbasis, a stronger condition.

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A Hermitian diagonalization example

$$S = \begin{bmatrix} 2 & i \\ -i & 2 \end{bmatrix} = S^H.$$

Characteristic polynomial:

$$\det(S - \lambda I) = (2-\lambda)^2 - |i|^2 = (\lambda - 1)(\lambda - 3).$$

Eigenvalues 1,3.

For $\lambda = 3$,

$$\begin{bmatrix} -1 & i \\ -i & -1 \end{bmatrix} z = 0$$

gives $z_1 = iz_2$. Choose

$$q_1 = (i, 1)^T / \sqrt{2}.$$

For $\lambda = 1$, choose

$$q_2 = (-i, 1)^T / \sqrt{2}.$$

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A Hermitian diagonalization example (continued)

Both vectors have unit norm and $q_1^H q_2 = 0$. Therefore

$$S = Q \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} Q^H,$$

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} i & -i \\ 1 & 1 \end{bmatrix}.$$

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The Fourier matrix

Let

$$\omega = e^{2\pi i/n}.$$

Forward Fourier matrix:

$$(F_n)_{jk} = \omega^{jk}, \qquad 0 \le j, k < n.$$

Forward transform:

$$y = F_n c.$$

The Fourier matrix is symmetric. It is generally not Hermitian for $n > 2$; sizes 1 and 2 are real symmetric exceptions. Its columns are orthogonal because

$$\sum_{j=0}^{n-1} (\omega^{jk})^H \omega^{j\ell} = \sum_j \omega^{j(\ell-k)} = n\delta_{k\ell}.$$

Therefore

$$F_n F_n^H = nI, \qquad F_n^{-1} = F_n^H/n.$$

Inverse DFT and Parseval

Inverse:

$$c_j = \frac{1}{n} \sum_{k=0}^{n-1} \omega^{-jk} y_k.$$

Software using

$$\omega = e^{-2\pi i/n},$$

interchanges the forward and inverse sign conventions.

Parseval:

$$\|y\|^2 = (F_n c)^H (F_n c) = n\|c\|^2.$$

Normalized unitary Fourier matrix:

$$Q = F_n/\sqrt{n}$$

satisfies $Q^H Q = I$, preserving the norm. Normalization changes scale constants without changing which information the transform represents.

DFT and cyclic convolution

Define cyclic convolution by $h_j = \sum_{k=0}^{n-1} a_k b_{(j-k) \bmod n}$. Its Fourier transform is

$$F_n h = (F_n a) \odot (F_n b)$$

Substitute this definition into $\sum_j \omega^{\ell j} h_j$ and set $s = (j-k) \bmod n$. The sum factors into $(\sum_k a_k \omega^{\ell k})(\sum_s b_s \omega^{\ell s})$. This identity uses the unnormalized forward transform. A forward transform divided by $\sqrt{n}$ requires an extra factor $\sqrt{n}$ on the right.

Two FFTs, pointwise multiplication, and an inverse FFT take $O(n \log n)$ arithmetic operations, compared with $O(n^2)$ for general direct cyclic convolution. Sequences of lengths p, q have linear convolution length $p + q - 1$; pad to at least that length to avoid wraparound.

Coefficient order, sign, and normalization must match the convolution definition. Record these conventions in FFT or finite-field transform implementations.

Even–odd decomposition in radix-2 FFT

Let $n = 2m$, $0 \le j < m$, and split the sequence into

$$c' = (c_0, c_2, \dots, c_{2m-2}),$$
$$c'' = (c_1, c_3, \dots, c_{2m-1}).$$

Separate $y_j = \sum_{k=0}^{n-1} c_k \omega_n^{jk}$ into even indices $k = 2s$ and odd indices $k = 2s+1$. Since $\omega_n^2 = \omega_m$, $\omega_n^m = -1$,

$$y_j = y'_j + \omega_n^j y''_j,$$
$$y_{j+m} = y'_j - \omega_n^j y''_j.$$

where y', y'' are DFTs of size $m = n/2$.

This butterfly operation combines two half-sized problems with m twiddle multiplications. Recursing to size 1 completes the FFT.

For $n = 4$, $c = (1, 2, 0, -1)^T$, bit reversal first gives

$$(c_0, c_2, c_1, c_3) = (1, 0, 2, -1).$$

Even–odd decomposition in radix-2 FFT (continued)

The first stage performs two two-point butterflies.

$$(1,0) \mapsto (1,1), \qquad (2,-1) \mapsto (1,3).$$

The second stage uses $\omega_4^0 = 1, \omega_4^1 = i$.

$$(1,1) \mapsto (2,0), \qquad (1,3) \mapsto (1+3i, 1-3i).$$

Consequently,

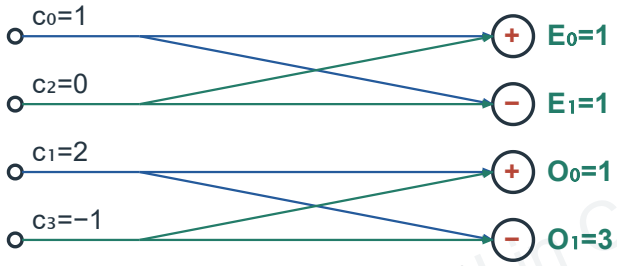
$$y = F_4 c = (2, 1+3i, 0, 1-3i)^T.$$

Direct evaluation gives $y_1 = 1 + 2i + (-1)(-i) = 1 + 3i$, $y_3 = 1 - 2i + (-1)i = 1 - 3i$. The plus and minus branches share the same twiddle product.

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Even–odd decomposition in radix-2 FFT (continued)

Four-point FFT stage 1: even and odd two-point DFTs



Even $(c_0, c_2) = (1, 0)$

$E_0 = c_0 + c_2$ · $E_1 = c_0 - c_2$

Odd $(c_1, c_3) = (2, -1)$

$O_0 = c_1 + c_3$ · $O_1 = c_1 - c_3$

Upper branch adds; lower subtracts green from blue. Next, pair matching E_j and O_j.

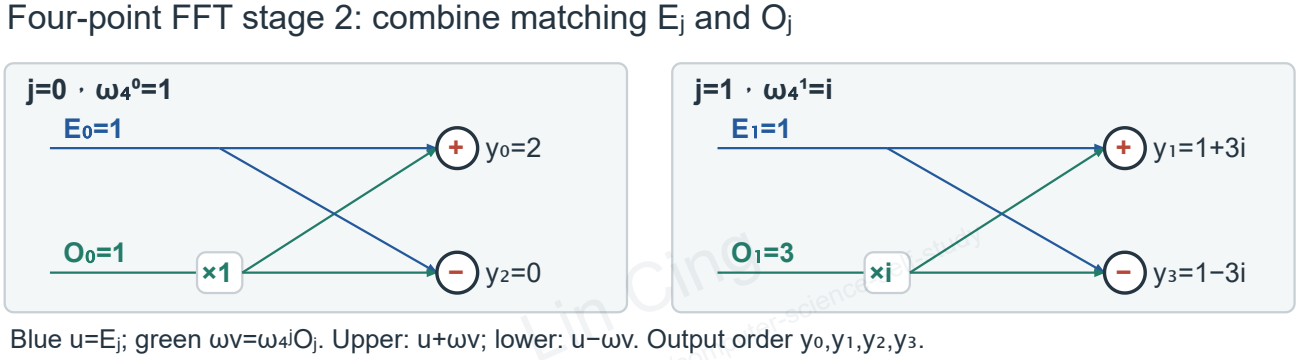
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Which intermediate values combine in a four-point FFT?

The first-stage array is $(E_0, E_1, O_0, O_1) = (1, 1, 1, 3)$. The second stage pairs positions $(0, 2)$ and $(1, 3)$. Only matching E_j, O_j produce y_j, y_{j+2}. The diagram separates the two butterflies to keep those pairings visible.

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Which intermediate values combine in a four-point FFT? (continued)



Blue $u = E_j$; green $\omega v = \omega_4^j O_j$. Upper: $u + \omega v$; lower: $u - \omega v$. Output order y_0, y_1, y_2, y_3.

The first three intermediate values happen to agree in this example, concealing a wrong pairing. Instead take $c = (2, 5, 7, -1)$. The first stage is $(9, -5, 4, 6)$, and the correct output is $(13, -5 + 6i, 5, -5 - 6i)$. Pairing E_0, E_1 incorrectly would give $9 + (-5) = 4$ instead of 13. Pairing indices are part of the algorithm.

Recursion and arithmetic cost

$n = 2^{\ell}$:

$$T(n) = 2T(n/2) + \Theta(n).$$

There are $\log_2 n$ levels with linear work at each level, giving

$$T(n) = \Theta(n \log n).$$

Naive Fourier matrix multiplication:

$$n^2$$

multiplications in the direct matrix formula.

For $n = 1024 = 2^{10}$,

$$n^2 = 1,048,576,$$
$$\frac{n}{2} \log_2 n = 5,120.$$

The theoretical multiplication-count ratio is 204.8 in this model. This is not a measured runtime speedup. Additions, memory access, parallel execution, and root generation affect actual runtime.

Iterative FFT and bit reversal

The recursion can be implemented bottom-up. A radix-2 decimation-in-time arrangement for $n = 8$ begins with bit-reversed input order

$$0, 4, 2, 6, 1, 5, 3, 7.$$

and applies butterflies of lengths 2, 4, and 8.

$$(a, b) \mapsto (a + \omega b, a - \omega b).$$

An inverse transform can conjugate the input, apply the forward transform, conjugate the output, and divide by n.

Other sizes can use mixed-radix methods or Bluestein's algorithm. A real-input FFT can exploit conjugate symmetry, but a generic complex interface need not do so automatically.

Numerical interpretation

Practical issues include the following.

- Compute or store twiddle factors accurately.
- Control accumulated phase error.
- Specify the forward scale and inverse sign.
- Zero-padding changes the frequency grid but creates no new observations.
- Windowing changes the signal while reducing some leakage.
- Bin frequency depends on the sampling interval through $k/(n\Delta t)$, with the signed-frequency interpretation fixed by convention.

A spectral peak can combine signal, leakage, and noise. A nonzero numerical bin alone does not establish an exact physical frequency.

Chapter check

You can now do the following.

- Compute complex products, moduli, powers, and roots.
- Use a consistent Hermitian inner product.
- Prove the Hermitian spectral properties.
- Construct a unitary eigenbasis.
- Check unitary norm preservation and eigenvalue moduli.
- Derive Fourier orthogonality and inversion.
- Implement radix-2 butterflies and derive $\Theta(n \log n)$ cost.
- Interpret signs, scaling, padding, and numerical limits.

Chapter 10 applies these structures to networks, engineering, Markov models, optimization, Fourier series, graphics, and cryptography.

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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